Welcome to Section 1

Introduction

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Course Overview

Section 1 thru 13

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Section 1 - Introduction

- Introduction to Database
- RDBMS
- What is SQL?
- Top 5 RDBMS
- Introduction to MS SQL Server
- Different editions of MS SQL

Section 2
Download, Install and Configure Windows

- MS SQL Server pre-requisite
- Lab setup
- What is virtualization?
- Download and Install VMWare Workstation Player
- Optional - Download and Install Oracle Virtualbox
- Create a Virtual Machine
- Download and Install Windows Server
- Virtual Machine Management

Section 3
Download, Install and Configure MS SQL Server

- Prerequisites for MSSQL Installation
- Download MSSQL and SQL Server Management Studio
- Install and Configure MSSQL
- Install SQL Server Management Studio
- Download and Install AdventureWorks Database

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Section 4
Database Fundamentals and Design

- What is Data and Database?
- How is data stored?
- What is a Table, COLUMN and ROW ?
- What is a key? primary, foreign, unique keys etc.
- What is Relational Database and Relational Database Management System (RDBMS) ?
- What is a Transaction and ACID properties?
- Database Normalization AND Different forms of Database Normalization
- Create your first Database, Table and Populate table with data…

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Section 5
Introduction to SQL Commands

- Welcome to Basic SQL Commands
- What is a SQL Statement and types of SQL statements
- DML Statement with examples
- DDL Statements with examples
- DCL Statement with examples
- TCL Statement with examples

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Section 6
Query and Manipulation of Data using SQL

- Create TABLE(s) and Temp Table(s)
- What Is a View?
- SELECT Statement in detail
- Operators, Expressions and Conditions
- WHERE Clause, ORDER BY, HAVING BY, GROUP BY Clause
- Select from two tables – JOINS
- Different Types of JOINS
- What is a Sub Query ?
- INSERT, UPDATE, DELETE and Truncate Statements
- What is a Store Procedure
- Function, Trigger and INDEX
- Clustered ,Non-clustered Indexes and Index Design considerations
- Index Fragmentation and Lab for Index

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Section 7
Microsoft SQL Database Administration

- Overview of MSSQL Management Tools
- Exploring SQL Server Management Studio (SSMS)
- Exploring SQL Server Configuration Manager
- MSSQL System Databases, DO's and DONT's of System databases
- What is master, etmpdb, msdb and other system databases?

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Section 8
Deep Dive into MSSQL Working

- What are Pages , Extents, Page Architecture and PFS?
- MSSQL Architecture
- MSSQL Database Architecture
- Operation and working of transaction log
- Configuring user Database
- Best practices while creating user Database

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Section 9
MSSQL Backup and Restore

- Backup of a Database, Importance for DBA and Different Media used for Backups
- Recovery Models in MSSQL and how that impact backups an restore
- Full and differential backups
- Transaction Log Backups, Log Backup Chain and Tail-log Backup
- Lab for Full, Differential and Transaction Log Backups
- Restore and Recovery Overview
- Full, Differential and Transaction Log database restore
- Point in time restore of a database
- Complete Backup , Restore and Restore in Time Lab
- MS SQL Server DBCC CHECKDB command
- Page Level Restore / Recovery using Full Backups
- Creating Maintenance Plan Backups / Re indexing etc.

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Section 10
MSSQL User Management

- MSSQL Security Model
- MSSQL Server Authentication Modes
- Server Logins
- Server RolesDatabase Users
- Database Roles
- Permissions (GRANT , DENY, and REVOKE)

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Section 11
MSSQL Server Agent Management

- SQL Server Agent Introduction
- SQL Server Agent Jobs and Schedules
- Agent Alerts
- Operators
- Database Mail
- Activity Monitor

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Section 12
Advanced SQL Server Administration Topics

- High Availability and its Types
- What Is Replication and Transactional Replication
- Transactional Replication Setup (LAB)
- MSSQL Data Encryption and type of encryptions
- Transparent Data Encryption (TDE) and configuration
- Backup and restore of encrypted Databases
- MS SQL Server Reporting Services (SSRS)
- MS SQL Server Integration Services (SSIS)

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Section 13
Course Recap

- Commands we have learned

Database

- Another component of IT or computers

Database

- Data is a collection of facts, such as numbers, words, measurements, observations or just descriptions of things. E.g. your name, age, height, weight, etc. are some data related to you. A picture, image, file, pdf, etc. can also be considered data.

- System
- Foundation
- Hub
- Central location

Database

A database is an organized and systematic collection of data generally stored and accessed electronically from a computer system. Databases make data management easy

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Database

- **Question**

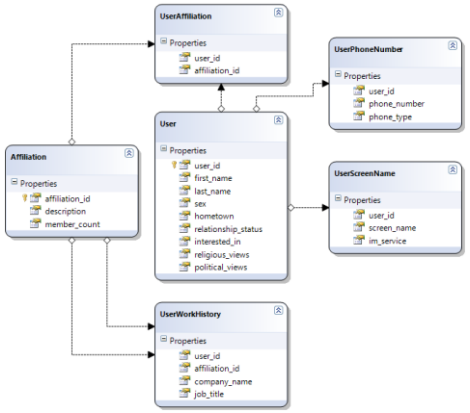
  Data can be stored in a spreadsheet or a text document as well so why we need a database?

- Databases store information more efficiently, databases can handle volumes of information that would be unmanageable in a spreadsheet.
- Spreadsheets have record limitations whereas databases do not
- Compared to databases, spreadsheets can require a large amount of hard-drive space for data storage
- Joining records is a lot easier in DB than compared to text or spreadsheets.

RDBMS

- RDBMS stands for relational database management system

- A relational database is a type of database. It uses a structure that allows us to identify and access data in relation to another piece of data in the database. Often, data in a relational database is organized into tables



Database

1	10001	Imran Afzal	IT
2	10002	Jerry Seinfeld	Finance
3	10003	George Costanza	Sales

10001	3 some place	917.000.1000
10002	10 Highrise ave	347.000.9000
10003	2020 Nosuch pl	212.000.0001

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RDBMS

RDBMS

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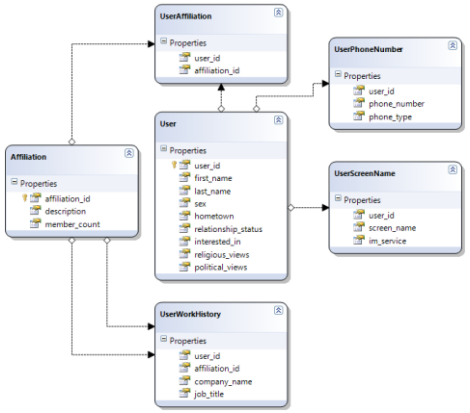

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What is SQL?

- SQL (**S**tructured **Q**uery **L**anguage) is a programming language used to communicate with data stored in a relational database management system. SQL syntax is similar to the English language, which makes it relatively easy to read, write, update or translate data

- SQL is often pronounced in one of two ways. You can pronounce it by speaking each letter individually like "**S Q L**", or pronounce it using the word "sequel".)

- It was invented in 1970s by IBM and the first version was called SEQUEL (**S**tructured **E**nglish **QUE**ry **L**anguage). Later it was changed to SQL because SEQUEL was already trademarked by an airline company

- SQL is NOT a database; it is a command line language that many RDBMSs use to access the data from tables

- Example statement: `SELECT * FROM tablename;`

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Top 5 RDBMS using SQL

- **Oracle DB**
  - It is a database product by Oracle and its enterprise version was released in 1979
  - It is a database commonly used for running online transaction processing, data warehousing and mixed database workloads
  - Oracle is one of the biggest vendor in the IT market

ORACLE
DB Management

- **Microsoft SQL Server**
  - MS SQL server is a relational database server by Microsoft
  - Its first version was released in 1988
  - It is among the most stable, secure and reliable database solutions
  - It supports wide variety of transaction processing, analytics, and business intelligence applications in corporate IT environments
  - MS SQL server competes primarily against Oracle DB and IBM DB2

Microsoft
SQL Server

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Top 5 RDBMS using SQL

- **MySQL**
  - A Swedish company called MySQL AB originally developed MySQL in 1994
  - It is community developed open source and its first version was release in 1995
  - The US tech company Sun Microsystems then took full ownership when they bought MySQL AB in 2008
  - US tech giant Oracle in 2009 acquired Sun Microsystems itself, and MySQL has been practically owned by Oracle since
  - Even though MySQL is open-source software, you can buy a commercial license version from Oracle to get premium support services
  - The main advantages of MySQL are that it is easy to use, inexpensive, reliable (has been around since 1995), and has a large community of developers who can help answer questions.

- **PostgreSQL**
  - Community developed open source and its first version was released in 1987
  - It was originally named POSTGRES, referring to its origins as a successor to the Ingres database developed at the University of California, Berkeley.
  - In 1996, the project was renamed to PostgreSQL to reflect its support for SQL
  - After a review in 2007, the development team decided to keep the name PostgreSQL.

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Top 5 RDBMS using SQL

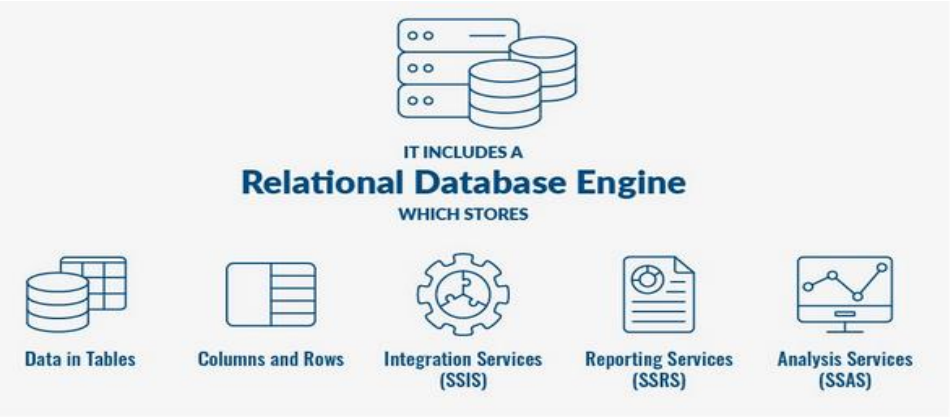
- **MariaDB**
  - Community developed open source and its first version was released in 2009
  - Development is led by some of the original developers of MySQL, who forked it due to concerns over its acquisition by Oracle Corporation in 2009
  - MariaDB is named after Monty's younger daughter, Maria

In this course our focus will be on:

Introduction to MS SQL Server

- MSSQL is a suite of database software published by Microsoft and used extensively in enterprise world

- It includes:
  - A relational database engine, which stores data in tables, columns and rows
  - Integration Services (SSIS), which is a data movement tool for importing, exporting and transforming data
  - Reporting Services (SSRS), which is used to create reports and serve reports to end users
  - Analysis Services (SSAS), which is a multidimensional database used to query data from the main database engine.



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Introduction to MS SQL Server

- In this course we will download, install and configure MS SQL 2016/2019 full enterprise evaluation version on Windows server 2016/2019 (180 days)
- Approximately 85% of corporate companies around the world uses MS SQL version 2016/2019 or older
- **SQL Server management studio** is an additional software that will be used to manage and maintain the databases, run SQL queries, perform backups and analyze performance charts

- Each version is named by the year but they are not exactly released in those specific years

- Most recent MS SQL versions are:
  - SQL Server 2014
  - SQL Server 2016
  - SQL Server 2017
  - SQL Server 2019.

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Introduction to MS SQL Server

Pros	Cons
Various supported editions	Expensive enterprise edition
Online product documentation	Difficult licensing process that's always changing
Microsoft Premier support	
On-premises and cloud database support	
Plenty of tools and applications	
Support for use on Linux	

IMPORTANT

<u>Please note:</u>

Regardless which RDBMS we choose our end goal is that you understand how to read, write and manipulate data using SQL statements.

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Editions of MS SQL Server

SQL Server Editions	Description
Enterprise	The premium offering, SQL Server Enterprise edition delivers comprehensive **high-end data center** capabilities with **blazing-fast performance**, unlimited virtualization, and end-to-end business intelligence—enabling high service levels for mission-critical workloads and end-user access to data insights
Standard	SQL Server Standard edition delivers **basic data management** and business intelligence database for departments and **small organizations** to run their applications and supports common development tools for on-premise and cloud—enabling effective database management with minimal IT resources.
Web	SQL Server Web edition is a **low total-cost-of-ownership** option for **web hosting companies**
Developer	SQL Server Developer edition lets developers build any kind of application on top of SQL Server. It includes **all the functionality of Enterprise** edition but is licensed for use as a **development and test system**, not as a production server. SQL Server Developer is an ideal choice for people who build and test applications
Express	Express edition is the **entry-level, free database** and is ideal for **learning and building desktop** and small server data-driven applications. It is the best choice for independent software vendors, developers, and hobbyists building client applications. If you need more advanced database features, SQL Server Express can be seamlessly upgraded to other higher end versions of SQL Server. SQL Server Express LocalDB is a lightweight version of Express that has all of its programming features. It runs in user mode and has a fast, zero-configuration installation and a short list of prerequisites

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